CHORUS: Checkpoint Orchestration with Rate-limiting and Unified Service-policies

Advanced Computing Project 25/26

André Miranda	PG60238
Gustavo Barros	PG61527
José Soares	PG60277
Miguel Carvalho	PG61153

Department of Informatics – School of Engineering – University of Minho

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Context & Problem

- **Context:** LLM training generates massive checkpoints (GBs to TBs).
- **The Bottleneck:** Parallel File Systems (PFS) like Lustre suffer from synchronized I/O bursts ("Noisy Neighbor" problem).
- **Consequence:** Compute nodes stall while waiting for slow synchronous writes, wasting expensive resources.

Central Insight

Checkpointing is **latency-critical for training continuation**, but not for long-term persistence.

System Components

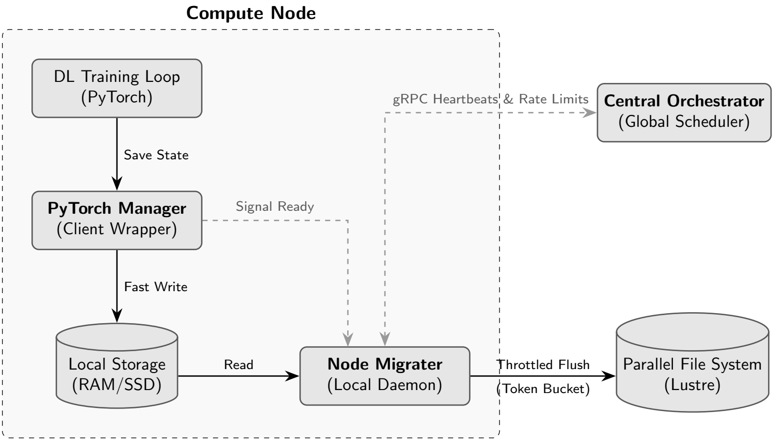


Figure 1. System architecture of the framework.

Two-Tier Storage Strategy

Local SSD Staging:

- Acts as a "Shock Absorber".
- Dedicated, uncontested bandwidth per node.
- Decouples training loop from network/PFS latency.

Predictability Result

Writing to local SSDs showed **near-zero variance** in stall times compared to the high volatility of direct PFS writes.

Pluggable Scheduling Policies

Policy	Logic
Uniform Fair Share	Splits bandwidth B_{PFS} equally among all N nodes, regardless of pending data.
Active Fair Share	Splits bandwidth B_{PFS} equally among nodes with pending data only.
Age Priority	Normalized combination of *pending size* and *waiting time* (anti-starvation).
Epoch Priority	Prioritizes jobs closer to completion (higher training progress ρ_i).

Dynamic Control: Orchestrator sends `CHANGE_RATE` or `HOLD` signals via persistent gRPC streams.

Token Bucket & Traffic Shaping

- **Mechanism:** Migrater read/write operations consume tokens refilled at the Orchestrator-assigned rate.
- **Chunk Size Tuning:** Evaluated trade-off between throughput accuracy and PFS syscall pressure.

Optimal Configuration

4 MiB chunks provided the best balance: high throughput accuracy with significantly reduced write frequency.

Results: Throughput Profiles

Burst Traffic

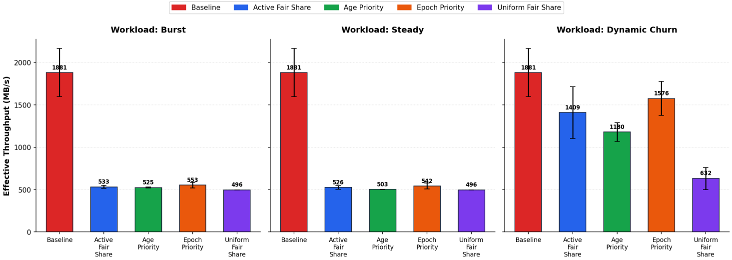
- Simultaneous dumps.

Steady Traffic

- Staggered intervals.

Dynamic Churn

- Job arrivals/departures.



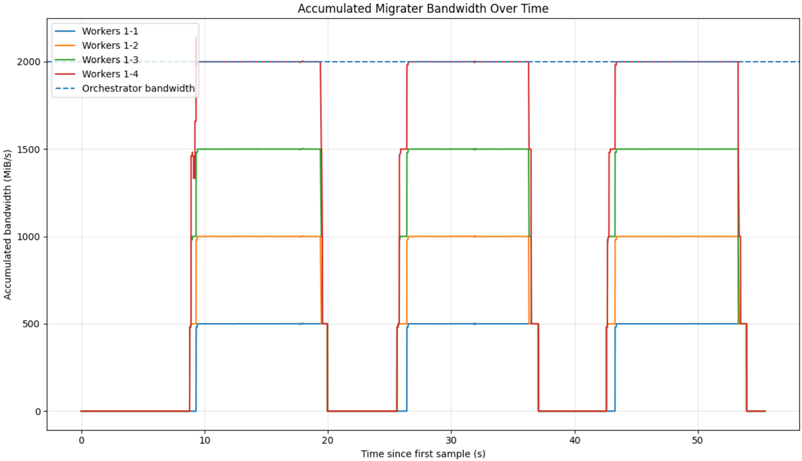
Setup Parameters:

- 5 GB checkpoint payload.
- 2 GB PFS bandwidth.

Cluster Environment:

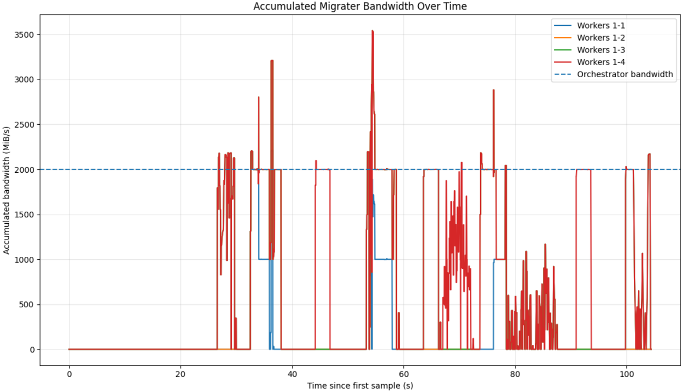
- Tracked concurrent SLURM jobs to account for shared network background noise.

Bandwidth Convergence



The system successfully transforms uncoordinated I/O bursts into smooth, regulated storage traffic. However...

Bandwidth Convergence (Issues)



- **Signaling Latency Layer:** Caused by the 0.1s gRPC heartbeat sampling stream.
- **Token Accumulation Spikes:** Empty initial bucket balance prevents initial bursts, but sub-second spikes remain at block boundaries.

Conclusions & Future Work

Conclusions:

- Two-tier orchestration provides **predictability** for LLM training.
- Centralized rate control ensures **Cluster Harmony**.
- Local SSD staging is a scalable option to absorb bursty I/O patterns.

Future Work:

- **Exascale:** Stress testing with thousands of concurrent nodes and weak scalability constraints.
- **Non-Blocking Queue Management:** Support queuing or pipelining new checkpoint requests if a worker generates a new state while the previous one is still migrating to the PFS.
- **Resilience:** Fault-tolerant Orchestrator (single point of failure).

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